2

Data Representation

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Key Concepts

1. Frequencies, cumulative frequencies, relative frequencies, and cumulative relative frequencies
2. Ungrouped and grouped frequency distributions
3. Sample size
4. Real limits and intervals
5. Frequency polygons
6. Normal, symmetric, and skewed frequency distributions
7. Percentiles, quartiles, and percentile ranks

In the first chapter we introduced the wonderful world of statistics. We discussed the value of statistics, met a few of the more well-known statisticians, and defined several basic statistical concepts, including population, parameter, sample, statistic, descriptive and inferential statistics, types of variables, and scales of measurement. In this chapter we begin our examination of descriptive statistics, which we previously defined as techniques that allow us to tabulate, summarize, and depict a collection of data in an abbreviated fashion. We used the example of collecting data from 100,000 graduate students on various characteristics (e.g., height, weight, sex, grade point average, aptitude test scores). Rather than having to carry around the entire collection of data in order to respond to questions, we mentioned that you could summarize the data in an abbreviated fashion through the use of tables and graphs. This way we could communicate features of the data through a few tables or figures without having to carry around the entire dataset.

This chapter deals with the details of the construction of tables and figures for purposes of describing data. Specifically, we first consider the following types of tables: frequency distributions (ungrouped and grouped), cumulative frequency distributions, relative frequency distributions, and cumulative relative frequency distributions. Next we look at the following types of figures: bar graphs, histograms, frequency polygons (or line graphs), cumulative frequency polygons, and stem-and-leaf displays. We also discuss common shapes of frequency distributions. Then we examine the use of percentiles, quartiles, percentile ranks, and box-and-whisker plots. Finally, we look at the use of SPSS and **R** and develop an APA-style paragraph of results. Concepts to be discussed include frequencies, cumulative frequencies, relative frequencies, and cumulative relative frequencies; ungrouped and grouped frequency distributions; sample size; real limits and intervals; frequency polygons; normal, symmetric, and skewed frequency distributions; and percentiles, quartiles and percentile ranks. Our objectives are that by the end of this chapter, you will be able to (a) construct and interpret statistical tables, (b) construct and interpret statistical graphs, and (c) determine and interpret percentile-related information.

2.1 Tabular Display of Distributions

Throughout this text, we will be following a group of superbly talented, creative, and energetic graduate research assistants (Challie Lenge, Ott Lier, Addie Venture, and Oso Wyse) working in their institution's statistics and research lab, fondly known as CASTLE

in an interval relative to the total sample. Thus, rather than stating that five individuals had scores in the 17 interval, we could say that 20% of the scores were in that interval. In the popular press, relative frequencies are quite often reported in tables without the frequencies (e.g., "56% of voters agreed with . . . "). Note that the sum of the relative frequencies should be 1.00 (or 100%) within rounding error. Also note that the *relative frequency distribution can be used with variables of all measurement scales,* from nominal (e.g., the percent of children with blue eye color) to ratio (e.g., the percent of adults who are 5'7").

2.1.4 Cumulative Relative Frequency Distributions

A fourth and final type of frequency distribution is known as the cumulative relative frequency distribution. For the example data this is depicted in the fifth column of Table 2.2 and labeled as "*crf* ." The number of **cumulative relative frequencies** (***crf***) for a particular interval *is the percentage of scores in that interval and smaller*. Thus, the 9 interval has a relative frequency of .04, and there are no relative frequencies smaller than that interval, so the cumulative relative frequency is simply .04. The 10 interval has a relative frequency of .04 and the relative frequencies less than that interval are .04, so the cumulative relative frequency is .08. The 11 interval has a relative frequency of .08 and the relative frequencies less than that interval total .08, so the cumulative relative frequency is .16. Thus, 16% of the people had scores in the 11 interval and smaller. In other words, *16% of people scored 11 or less*. One way to think about determining the cumulative relative frequency column is to take the relative frequency column and accumulate *downward* (i.e., from the top down, yielding .04; .04 + .04 = .08; .04 + .04 + .08 = .16; etc.). Just as a check, the *crf* in the largest interval should be equal to 1.0, within rounding error, just as the sum of the relative frequencies is equal to 1.0. Also note that the cumulative relative frequency distribution can be used with variables of measurement scales from ordinal (e.g., the percent of students receiving a B or less) to interval and ratio (e.g., the percent of adults who are 5'7" or shorter). As with relative frequency distributions, cumulative relative frequency distributions cannot be used with nominal data.

2.2 Graphical Display of Distributions

In this section we consider several types of graphs for viewing a distribution of scores. Again, we are still interested in how the data for a single variable can be represented, but now in a graphical display rather than a tabular display. The methods described here include the bar graph; histogram; frequency, relative frequency, cumulative frequency and cumulative relative frequency polygons (or line graphs); and stem-and-leaf display. Common shapes of distributions will also be discussed.

2.2.1 Bar Graph

A popular method used for displaying nominal scale data in graphical form is the **bar graph**. As an example, say that we have data on the eye color of a sample of 20 children. Ten children are blue-eyed, six are brown-eyed, three are green-eyed, and one is black-eyed. Note that this is a *discrete* variable rather than a continuous variable. A bar graph for

2.3 Percentiles

In this section we consider several concepts and the necessary computations for the area of percentiles, including percentiles, quartiles, percentile ranks, and the box-and-whisker plot. For instance, you might be interested in determining what percentage of the distribution of the GRE-Quantitative subtest fell below a score of 165 or in what score divides the distribution of the GRE-Quantitative subtest into two equal halves.

2.3.1 Percentiles

Let us define a **percentile** as that score below which a certain percentage of the distribution lies. For instance, you may be interested in that score below which 50% of the distribution of the GRE-Quantitative subscale lies. Say that this score is computed as 150; this would mean that 50% of the scores fell below a score of 150. Because percentiles are scores, they are continuous values, and can take on any value of those possible. The 30th percentile could be, for example, the score of 145. For notational purposes, a percentile will be known as P_i, where the i subscript denotes the particular percentile of interest, between 0 and 100. Thus the 30th percentile for the previous example would be denoted as $P_{30} = 145$.

Let us now consider how percentiles are computed. The formula for computing the P_i percentile is

$$P_i = LRL + \left(\frac{(i\%)(n) - cf}{f} \right)(w)$$

where *LRL* is the lower real limit of the interval containing P_i, i% is the percentile desired (expressed as a proportion from 0 to 1), n is the sample size, *cf* is the cumulative frequency less than but not including the interval containing P_i (known as "*cf* below"), f is the frequency of the interval containing P_i, and w is the interval width.

As an example, consider computing the 25th percentile of our statistics quiz data. This would correspond to that score below which 25% of the distribution falls. For the example data in the form presented in Table 2.2, we can compute P_{25} as follows:

$$P_i = LRL + \left(\frac{(i\%)(n) - cf}{f} \right)(w)$$

$$P_{25} = 12.5 + \left(\frac{(25\%)(25) - 5}{2} \right)(1) = 12.5 + 0.625 = 13.125$$

Conceptually, let us discuss how the equation works. First we have to determine what interval contains the percentile of interest. This is easily done by looking in the *crf* column of the frequency distribution for the interval that contains a *crf* of .25 somewhere within the interval. We see that for the 13 interval the *crf* = .28, which means that the interval spans a *crf* of .20 (the *URL* of the 12 interval) up to .28 (the *URL* of the 13 interval), and thus contains .25. The next largest interval of 14 takes us from a *crf* of .28 up to a *crf* of .32, and thus is too

2.4 Recommendations Based on Measurement Scale

We cannot stress enough how important it is that you understand the measurement scale of the variable(s) with which you are working, as that will dictate what statistics can (and cannot) be computed using them. You will use the knowledge of measurement scale in every statistic that you compute. To help in this endeavor, we include Box 2.1 as a summary of which data representation techniques are most appropriate for each type of measurement scale.

BOX 2.1 Appropriate Data Representation Techniques Given the Measurement Scale of the Variable

Measurement Scale	Tables	Figures
Nominal	Frequency distribution	Bar graph
	Relative frequency distribution	
Ordinal, interval, or ratio	Frequency distribution	Histogram
	Cumulative frequency distribution	Frequency polygon
	Relative frequency distribution	Relative frequency polygon
	Cumulative relative frequency distribution	Cumulative frequency polygon
		Cumulative relative frequency polygon
		Stem-and-leaf display
		Box-and-whisker plot

2.5 Computing Tables, Graphs, and More Using SPSS

The purpose of this section is to briefly consider applications of SPSS for the topics covered in this chapter (including important screenshots). We will begin with a brief introduction to SPSS and then demonstrate SPSS procedures for generating frequencies and graphs.

2.5.1 Introduction to SPSS

Before we get into using SPSS, let's go over a few basics. SPSS is one of the most common standard statistical software programs available. Among other nice features of SPSS is that it is user-friendly, particularly compared to many other standard statistical software. Most SPSS users take advantage of the point-and-click interface of SPSS, although you can also use syntax to run statistics in SPSS. While the graphical user interface makes generating statistics in SPSS pretty easy, you need to be aware of a few nuances when working with the program.

Graphs can also be generated in SPSS using Chart Builder (click on "Graphs" in the top pulldown menu and then go to "Chart Builder"). This is an interactive tool that allows researchers to drag and drop variables and types of graphs (e.g., bar, line, boxplots, and more) to build the figure. On the right side, "Element Properties" and "Chart Appearance" provide researchers with aesthetic options. Should you not choose to use Chart Builder to create your graph, you still have the ability to adapt the look of your graph by double-clicking on the graph in your output. This will allow you to access Chart Editor and make aesthetic alterations to your figure.

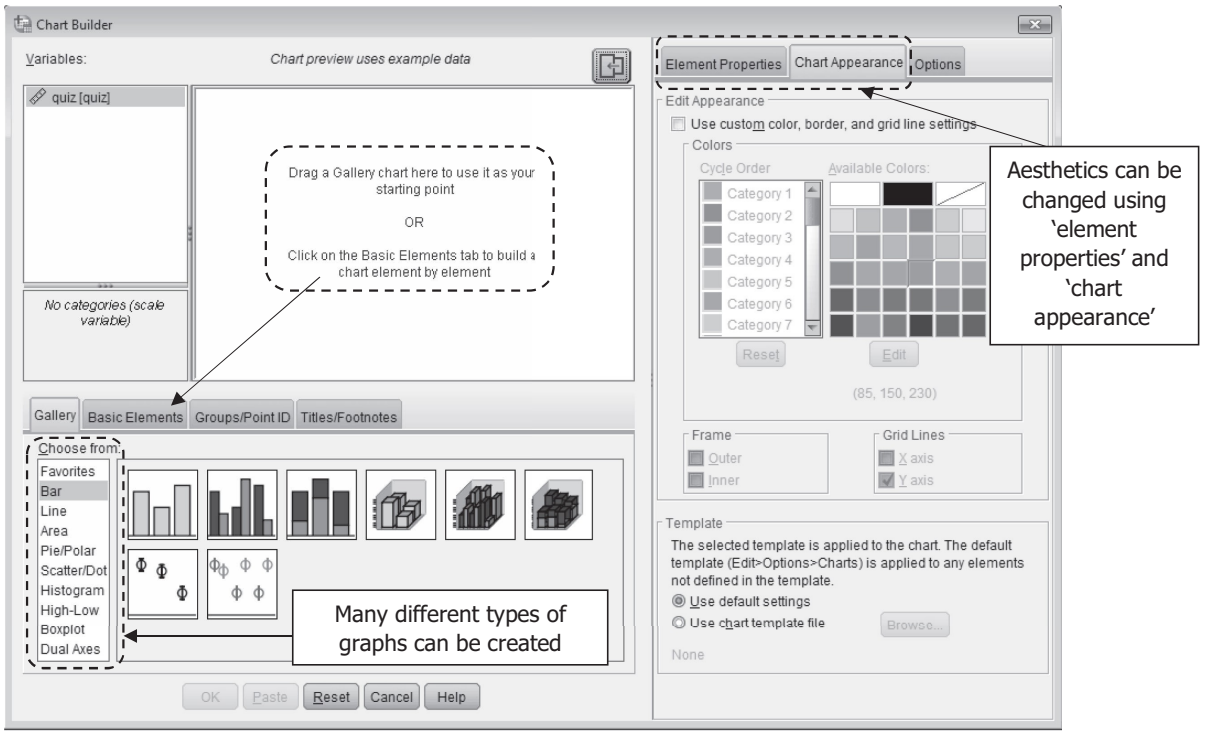


FIGURE 2.23
Chart Builder.

2.6 Computing Tables, Graphs, and More Using R

2.6.1 Introduction to R

The purpose of this section is to briefly consider applications of **R** for the topics covered in this chapter. We will begin with a brief introduction to **R** and then describe **R** procedures for frequencies and graphs.

R is a freely accessible open source software environment that operates on both Windows and Mac platforms. *Open source* means that anyone can contribute to the environment, and thus a plethora of exciting tools have been, and will continue to be, developed in **R**. We will interject here that we have a love–hate relationship with **R**. By being free and open source, **R** is broad and deep in terms of the tools available, with more being added almost daily. Additionally, the **R** community is unparalleled for the help and support offered to users. On the other hand, **R** does not operate in a point-and-click environment. Rather, **R** operates using command language and users have to write "scripts" (i.e., code or syntax) to *tell* **R** what to do, and **R** is very finicky in its prose. At this point, you may be asking: "Why in

2.7 Research Question Template and Example Write-Up

Depending on the purpose of your research study, you may or may not write a research question that corresponds to your descriptive statistics. If the end result of your research paper is to present results from inferential statistics, it may be that your research questions correspond only to those inferential questions, and thus no question is presented to represent the descriptive statistics. That is quite common. On the other hand, if the ultimate purpose of your research study is purely descriptive in nature, then writing one or more research questions that correspond to the descriptive statistics is not only entirely appropriate, but (in most cases) absolutely necessary. At this time, let us revisit our graduate research assistant, Addie Venture, who was introduced at the beginning of the chapter. As you may recall, her task was to summarize data from 25 students enrolled in Dr. Debhard's statistics course. The questions that Addie developed based on consultation with Dr. Debhard included the following:

1. What interpretations can be made from the frequency table of quiz scores from students enrolled in an introductory statistics class?
2. What interpretations can be made from graphical representations of quiz scores from students enrolled in an introductory statistics class?
3. What is the distributional shape of the statistics quiz scores?
4. What is the 50th percentile of the quiz scores?

A template for writing descriptive research questions for summarizing data follows. Note that these are just a few examples. Given the multitude of descriptive statistics that can be generated, these are not meant to be exhaustive:

What interpretations can be made from the table of [variable]? What interpretations can be made from the graphical representation of [variable]? What is the distributional shape of the [variable]? What is the 50th percentile of [variable]?

Next, we present an APA-like paragraph summarizing the results of the statistics quiz data example:

As shown in Table 2.2 and Figure 2.2, scores ranged from 9 to 20, with more students achieving a score of 17 than any other score (20%). From Figure 2.2 we also know that the distribution of scores was negatively skewed, with the bulk of the scores being at the high end of the distribution. Skewness was also evident as the quartiles were not equally spaced, as shown in Figure 2.7. Thus, overall the sample of students tended to do rather well on this particular quiz, although a few low scores should be troubling (as 20% did not pass the quiz and suggest the need for some remediation).

2.8 Additional Resources

Throughout the chapter, we've shared a number of resources related to graphing. As you step into learning statistical software, a number of excellent tools are available to help you:

- An increasing number of books are available for learning **R** (e.g., Crawley, 2013; Rahlf, 2017; Wickham & Grolemund, 2017). Wickham and Grolemund (2017) is also available online at https://r4ds.had.co.nz.
- As mentioned in the text, the **R** user community is both large and immensely helpful. The following websites are a few places you can start when you need **R** support:
  - **R** cookbook: www.cookbook-r.com
  - Datacamp: www.datacamp.com
  - Stackoverflow: https://stackoverflow.com/questions/tagged/r
  - Comprehensive **R** Archive Network (CRAN): https://cran.r-project.org
  - Learning **R** in **R**: https://swirlstats.com
- Discussion lists are one of the quickest ways to find support; users of SPSS can join the SPSS listserv to post questions or search the archives (https://listserv.uga.edu/cgi-bin/wa?A0=SPSSX-L).

Problems

Conceptual Problems

1. For a distribution where the 50th percentile is 100, what is the percentile rank of 100?

a. 0

b. .50

c. 50

d. 100

2. Which of the following frequency distributions will generate the same relative frequency distribution?

X	*f*	*Y*	*f*	*Z*	*f*
100	2	100	6	100	8
99	5	99	15	99	18
98	8	98	24	98	28
97	5	97	15	97	18
96	2	96	6	96	8

a. *X* and *Y* only

b. *X* and *Z* only

c. *Y* and *Z* only

d. *X*, *Y*, and *Z*

e. None of the above